

Representation Collapse Is Not the Bottleneck: A Negative Result and Diagnosis for Vision-Based Quadrotor Hover

Draft v0.4 — 2026-06-23. Target: ICRA / robot-learning workshop (venue TBD). **Simulation-only; no real-robot claim.** This draft is assembled from the frozen-protocol leaderboard and the Phase 3 diagnostic reports; every number is reproducible from the cited script/artifact. **v0.4 re-syncs the core ablation (§5) and conclusion (§9) to the official-code-faithful Dispersive re-run (P2f, InfoNCE-L2 on the `flow_net` mid-block, $\lambda=0.5$, $\tau=0.5$, with the `/d` per-dimension normalisation): the negative result is confirmed and the earlier "byte-identical no-op" control is overturned (§5).**

Abstract

Diffusion / flow-matching visual policies are prone to *representation collapse* when fine-tuned end-to-end, and **Dispersive Loss** — a contrastive regulariser that repels intermediate features — has been proposed to prevent it and thereby improve high-frequency visual control. We pre-registered and tested this hypothesis on a vision-based quadrotor hover task (monocular 64×64 FPV + IMU, flow-matching policy, 50 Hz closed loop). Under a **frozen evaluation protocol** (paired initial conditions, conditional-on-survival precision, bootstrap CIs, across-seed mean $\pm$ std, a *measured* state-based oracle), a **2×2 ablation** (Dispersive $\times$ end-to-end encoder, three seeds per cell) — run with an **official-code-faithful** Dispersive Loss (InfoNCE-L2 on the generative `flow_net` mid-block, $\lambda=0.5$, $\tau=0.5$, including the `/d` per-dimension normalisation) — finds **no survival or task-precision gain above seed noise** (−2.2 pp Tier-1 within a 6.3 pp pooled std; survival −2.1 pp). Because the implementation follows the released code exactly — placement, weight, and per-dimension normalisation — the null cannot be attributed to a fidelity error. The frozen-encoder condition is an informative control rather than a no-op: under the faithful mid-block placement the term trains the generative network **even with a frozen encoder**, where it **destabilises** control (mean Tier-1 87.8→74.4, variance up) rather than leaving it inert. Reaching even this null verdict required a protocol hardened against **single-seed noise**: our own from-pixels PPO baseline swings between a 0 % and 47 % Tier-1 pass-rate on the training seed alone, so we report every retrained model as an **across-seed mean $\pm$ std** rather than a single leaderboard row. We then diagnose *why*: (i) Dispersive does not cure collapse — at **both** the faithful `flow_net` mid-block and an off-path placement (§6.1) it **games its own objective**, inflating feature norm ($\sim 9\times$ and $\sim 287\times$ respectively) while the intrinsic rank gets *worse*, so closed-loop survival is **decoupled** from that geometry (a large rank swing leaves survival flat); (ii) the only real survival mover is a transfer/conditioning/recovery **recipe**, not Dispersive; (iii) precision is gated neither by representation nor by sensing. Although the 64×64 FPV cannot encode metric range beyond ~ 2 m, a renderer gate shows that loss is a fixable target artifact (a non-saturating target restores it at 64 px), and a positive-control **intervention** settles it: *handing the policy the oracle metric position error barely moves precision* ($\sim 36\times$ the state oracle) and a richer cue **collapses survival** — so precision is capped by the absence of a learned far-range recovery behaviour (a coverage / teacher-competence gap), not by the observation channel. We conclude that for this class of task neither representation collapse nor sensing is the binding constraint. We release the protocol, the intervention, and all diagnostics.

1. Introduction

End-to-end visual control with generative policies (diffusion [3], flow matching [4]) has advanced rapidly, but a recurring failure mode is *representation collapse*: when the visual encoder is trained jointly with the action head, its features can degenerate onto a low-dimensional subspace, losing the information the controller needs. **Dispersive Loss** [13] addresses this by adding a contrastive term that repels a batch's intermediate features, and has been reported to improve generative-policy control [14].

We asked a simple, falsifiable question on a concrete task — **vision-based quadrotor hover** — does Dispersive Loss improve closed-loop control, and does it need an end-to-end-trainable encoder to do so? The task is a good stress test: the policy runs at 50 Hz from a monocular 64×64 FPV stream plus IMU, with a flow-matching action head; covariate shift and feature collapse are both plausible failure modes.

Our answer is **no**, and the value of the paper is the *diagnosis* of why. We make three contributions:

1. **A frozen evaluation protocol** for visual hover that is robust to the short-survival *and* single-seed artifacts that silently mislead precision metrics and leaderboard rows for policies that crash early or train unstably (paired seeds, conditional-on-survival precision, bootstrap CIs, across-seed mean $\pm$ std, and a *measured* rather than assumed oracle).

2. **A pre-registered 2×2 falsification** (three seeds per cell) of the Dispersive-Loss hypothesis, hardened with an **official-code-faithful** implementation (InfoNCE-L2 on the `flow_net` mid-block, $\lambda=0.5$, `/d`), so the null result controls for implementation fidelity, plus a frozen-encoder control.
3. **A mechanistic diagnosis** separating the three candidate bottlenecks (representation, data coverage, sensing) and localising the real one — including a positive-control *intervention* that hands the policy the oracle sensing signal and shows it does **not** restore precision, isolating the teacher's far-range incompetence as the binding constraint.

This is a negative result, but a constructive one: it redirects effort from representation regularisation to the observation model. Because the regulariser is implemented exactly as released — placement, weight, and per-dimension normalisation (§3) — the null result cannot be attributed to an implementation error.

2. Related Work

Dispersive regularization at a transfer boundary: from manipulation to high-speed visual hover.

Generative policies for visuomotor control. Behaviour cloning treats control as supervised regression from observations to actions, but robot action distributions are multimodal, sequentially correlated, and precision-sensitive, which makes plain regression brittle. Building on denoising diffusion [1] and its deterministic DDIM sampler [2], Diffusion Policy [3] reframed control as conditional denoising diffusion over short action sequences, with receding-horizon execution and a GroupNorm visual encoder trained end-to-end. Flow matching [4] and rectified flow [5] offer a continuous-time, simulation-free alternative that learns a velocity field from noise to data, since adopted as a lighter-weight policy backbone — e.g. the π_0 vision-language-action flow model [6] and the Riemannian flow-matching policy [7]. Our policy is a flow-matching action head over a fused vision-IMU conditioning vector, placing it in this family.

Because imitation alone cannot exceed its demonstrations, a second line fine-tunes pretrained generative policies with reinforcement learning. DPPO [8] casts the denoising chain as a multi-step MDP and applies PPO; ReinFlow [9] injects a learnable noise network into a flow policy's otherwise deterministic path, recovering a tractable likelihood that supports stable RL fine-tuning at very few — or even one — denoising steps; Q-score matching [12] is a related value-based route. This interactive, closed-loop setting should be distinguished from RL fine-tuning of *image-generation* diffusion models (DDPO [10]; DPOK [11]), which optimise a non-interactive sampler against a reward. This generative-policy-plus-RL stack is the substrate on which the regularizer studied here is layered, and the same stack in which the positive results for that regularizer were reported.

Representation regularization in generative models and policies. A recurring concern in denoising-based generators is that regression objectives supervise the output but leave intermediate features under-constrained, allowing them to collapse. The failure has a well-studied form: Jing et al. [22] show that even contrastive objectives suffer *dimensional collapse*, with embeddings spanning a lower-dimensional subspace than their nominal dimension — the property later diagnostics quantify through effective rank. REPA [15] addresses this by *aligning* a diffusion transformer's hidden states with a pretrained self-supervised encoder, at the cost of an external encoder and data; the classic variance/covariance and redundancy-reduction criteria of VICReg [16] and Barlow Twins [17] pursue the same anti-collapse goal without positive pairs. Dispersive Loss [13] removes the external-encoder cost differently: it keeps only the *repulsive* half of a contrastive objective, encouraging a generative network's intermediate representations to spread out with no positive pairs, no extra parameters, and no external data, validated on ImageNet class-conditional generation where it consistently improves FID.

Two robotics works carry Dispersive Loss into control. D²PPO [14] identifies "diffusion representation collapse" — semantically similar observations mapped to indistinguishable features — as a cause of manipulation failures, adds dispersive regularization treating every hidden representation in a batch as a negative pair, and reports average gains of 22.7 % in pre-training and 26.1 % after PPO fine-tuning on RoboMimic, with high real-robot success on a Franka Panda. DMPO [23] extends the idea to one-step MeanFlow policies, arguing dispersive regularization is necessary for stable single-step generation; concurrent MeanFlow-plus-dispersive manipulation work includes DM1 [18] and MP1 [19]. Notably, the public DMPO configuration specifies InfoNCE-L2 on the mid-block with weight 0.5 — the settings a faithful re-implementation in a new domain would inherit, and the settings we adopt in §3 / §5.

Three properties of this literature matter here. First, every validation is on image generation [13] or quasi-static, table-top manipulation scored by discrete task success [14, 23]. Second, the mechanism on offer is *observational discriminability*: the failure dispersive loss is claimed to fix is confusing two similar images. Third, none of these works publishes a fidelity audit — placement, weight, the per-dimension normalization in the official InfoNCE-L2 form — or a frozen-encoder control that isolates whether the term is even on the action-gradient path. Whether the benefit, the mechanism, or even the on-path behaviour transfers to a setting with a different bottleneck is left open.

Learning-based vision quadrotor control. High-speed visual flight is exactly such a setting. The dominant recipe couples privileged-learning imitation with sensor abstraction: Deep Drone Racing [24] maps images to a waypoint and desired speed for a downstream planner; Deep Drone Acrobatics [25] regresses collective thrust and body rates from abstracted visual-inertial inputs to fly 3g maneuvers by imitating a privileged expert; Learning High-Speed Flight in the Wild [20] maps depth and IMU to collision-free trajectories with zero-shot sim-to-real transfer; and champion-level drone racing was achieved with deep RL from onboard perception [21]. A consistent theme is that what unlocks robustness and transfer is *data diversity, sensor abstraction, and privileged supervision* — not auxiliary representation regularizers. The failure mode here is also different in kind: a hover or recovery policy does not fail by confusing two similar frames, it fails by drifting off-distribution and crashing, the classic covariate-shift problem of behaviour cloning [26].

Negative results, auxiliary-loss (in)effectiveness, and honest evaluation. Representation-centric add-ons do not help uniformly. Most directly, Schneider et al. [27] benchmark pretrained visual representations in model-based RL and find — "surprisingly" — that they are no more sample-efficient and no more OOD-robust than features learned from scratch, concluding that data diversity and architecture, not representation quality, drive OOD generalization. Establishing a credible negative result also depends on evaluation discipline: Henderson et al. [28] document how seed variance makes single-run RL claims unreliable; Agarwal et al. [29] show that few-run benchmarks are statistically fragile and advocate bootstrap intervals; Patterson et al. [30] catalogue maximization bias and unfair hyperparameter budgets. Together they motivate the frozen, multi-seed protocol with interval estimates (§4) on which our null effect is asserted rather than assumed.

Positioning: a transfer boundary, not a contradiction. We take Dispersive Loss following the official implementation — InfoNCE-L2 on the generative network's mid-block, weight 0.5, including the per-dimension normalization present in the released code — and ask whether the benefit reported for image generation [13] and table-top manipulation [14, 23] transfers to high-speed visual quadrotor hover. Under a frozen, three-seed protocol with bootstrap intervals (following [28, 29]), it does not: against D²PPO's reported +22.7 % / +26.1 % manipulation-success gains on RoboMimic [14], we measure a −2.2 pp Tier-1 change that sits inside a 6.3 pp across-seed std, with survival and precision unmoved (§5–§6). This is best read not as a contradiction of the manipulation results but as a *transfer boundary*. Where the bottleneck is observational discriminability — telling two near-identical grasp scenes apart — spreading intermediate features can help [14]. Where the bottleneck is behavioural coverage — recovering from a one-to-three-metre offset before crashing — feature dispersion addresses the wrong axis, echoing Schneider et al.'s [27] finding that representation quality is not the constraint governing out-of-distribution behaviour. Our contribution is a faithful, pre-registered, multi-seed demonstration of where a popular regularizer stops working on this task, and a diagnosis of why.

3. Task, Policy, and the Dispersive Mechanism

Task. Hover a quadrotor at a target. The simulator is a 6-DOF rigid body (200 Hz RK4 inner loop, INDI attitude control) with a 50 Hz outer control loop. The agent observes a 2-frame stack of 64×64 RGB synthetic FPV images and a 6-D IMU vector (gyro + specific force); it outputs a short sequence of collective-thrust/body-rate (CTBR) commands, executing the first and re-observing (receding horizon).

Policy (flow matching, v5). A CNN vision encoder produces a pooled feature $\text{vis_pooled} \in \mathbb{R}^{256}$ and a spatial map $\mathbb{R}^{256 \times 4 \times 4}$. An IMU MLP produces $\text{imu_feat} \in \mathbb{R}^{512}$. An IMU-to-vision cross-attention (IMU query, spatial keys/values) produces $\text{attended} \in \mathbb{R}^{256}$. The flow network conditions on $\text{global_cond} = [\text{attended}; \text{imu_feat}] \in \mathbb{R}^{768}$. A training-only auxiliary head predicts the 15-D privileged state from vis_pooled .

Dispersive Loss (faithful, official recipe). We apply Dispersive Loss exactly as released in the official code [13, 14]: the InfoNCE-L2 form on the generative network's intermediate block — here the `flow_net` mid-block (`return_mid=True`), $\mathcal{L}_{\text{disp}} = \log E_{i \neq j} [\exp(-D/\tau)]$, $D = \|z_i - z_j\|^2 / d$ — with weight $\lambda=0.5$ and $\tau=0.5$, including the per-dimension $/d$ normalization present in the released code. (Implementation: `models/flow_policy_v5.py:204–224, 269–287`.) Because the term lands on `flow_net` mid-features, its gradient reaches the action path **regardless of whether the visual encoder is trainable** — the property that makes the frozen-encoder control in §5 informative.

Fidelity of the implementation, and an off-path variant studied in §6.1. Two configuration choices determine whether the term is active at all, and we fix both to the released values. *Placement, weight, and form:* the term must sit on a generative-network intermediate block (here the `flow_net` mid-features) at $\lambda=0.5$ in the InfoNCE-L2 form. A hand-rolled log-distance variant $-\text{mean}_{i \neq j} \log(\|x_i - x_j\| + \epsilon)$ at $\lambda=0.05$ applied to vis_pooled — a feature that feeds **only** the auxiliary state head, not the action path (which conditions on `attended` and `imu_feat`) — departs from the released recipe on all three axes. *Normalisation:* the published Algorithm 1 omits the per-dimension $/d$ divisor present in the released code; without it the loss saturates to zero gradient at $\tau=0.5$, so this divisor is required for the term to carry gradient at all. The headline ablation (§5) uses the released configuration on both axes. Separately, §6.1 studies the off-path vis_pooled placement as a mechanistic probe, where its off-path position exposes how a scale-sensitive dispersion term can game its objective.

Oracle. A state-based PPO controller reading the privileged 15-D state is the performance upper bound.

4. Frozen Evaluation Protocol

The project's metric had changed three times (RMSE → linear-clip hierarchical → exp-decay hierarchical), making cross-run numbers incomparable; a **short-survival artifact** kept inflating precision for policies that crash early (a policy that dies at step 24 was credited with a "precise" 0.6 m hover from its few alive steps); and, as we demonstrate below with our own PPO baseline, **single-seed evaluation is itself unreliable** — a collapse-prone policy can swing from Tier-1 0 % to 47 % on the training seed alone. We freeze one protocol (`evaluate_frozen_p0.py`) and do not change it:

- **Paired initial conditions.** Episode i uses `seed = 12345 + i` to seed the env, the global RNG (visual domain randomisation), and torch (flow noise), so every model sees byte-identical initial conditions — a paired comparison.
- **Primary axes: survival and Tier-1 pass-rate** (fraction of 30 episodes flying $\geq 250/500$ steps). These are robust to the short-survival artifact.
- **Conditional precision.** Integrated absolute error (IAE) of position is reported **only over episodes that survived ≥ 250 steps** (`cond-IAE` , with its support `n_cond`); the naive all-episode IAE is reported but flagged.
- **Bootstrap 95% CIs** over episodes.
- **Across-seed mean $\pm$ std** for every model we *retrain* in this work — Table 1's P1 baselines and Table 2's 2×2 cells — over three training seeds. The legacy flow-policy checkpoints predate this work and are single-seed; that is a stated limitation, and the seed sensitivity we document below is exactly why a single training run is not a safe basis for a leaderboard row.
- **Measured oracle.** The state-based PPO oracle is rolled through the *same* protocol; its composite score (**0.9668**, 100 % survival, 0.068 m hover IAE) is the normaliser (`%Oracle`), replacing a previously hard-coded constant.

Under this protocol, **three single-seed claims fail to survive**. Two come from the prior leaderboard: "H4 = SOTA" (true only on a multiplicative composite that rewards *die-early-but-precise*; H4 survives 40.8 %, Tier-1 13.3 %) and "RL gives 51 % precision gain" (a short-survival artifact: its conditional IAE is the *worst* of the group, 3.34 m, on `n_cond` = 4/30). The third we generate ourselves and catch with the seed protocol: on a single seed our PPO-from-pixels baseline "uniformly collapses" (Tier-1 0 %), but over three seeds it is only *collapse-prone* — a high-variance Tier-1 of **15.6 $\pm$ 22.0 %**, with one seed reaching 47 % — so the single-seed verdict simultaneously over- and under-states the behaviour. The lesson is structural, not incidental: on this task one training seed is not a safe basis for a leaderboard row, which is why every model we retrain is reported as a seed mean $\pm$ std. **No vision policy exceeds $\sim$ 17 % of oracle; none is deployable.** (Table 1.)

Table 1 — Frozen leaderboard (flow policies: `evaluation_results/frozen_p0_leaderboard.json` ; P1 baselines ^B, 3 seeds: `evaluation_results/baselines_frozen_leaderboard.json` , `evaluation_results/baselines_frozen_ppopx.json` , `per-seed ..._bc,ppopx}_s{1,2}.json` , aggregate `...seeds_aggregate.json`).

Model	Score	Survive	Tier-1	cond-IAE (n)	%Oracle
PPO Oracle (state)	0.967 [.966,.967]	100 %	100 %	0.068 m (30)	100 %
H4_BC	0.168 [.154,.182]	40.8 %	13.3 %	2.520 m (4) Δ	17.4 %
v5_RL_best	0.129 [.117,.139]	28.8 %	13.3 %	3.340 m (4) Δ	13.3 %
v5_BC	0.124 [.107,.143]	55.9 %	70.0 %	2.724 m (21)	12.8 %
PPO-from-pixels ^B	0.114 $\pm$ 0.017	30.1 $\pm$ 15.6 %	15.6 $\pm$ 22.0 %	2.83 m (1/3) $\Delta\Delta$	11.8 $\pm$ 1.8 %
Joint_E2E_v5	0.106 [.095,.118]	62.2 %	80.0 %	3.077 m (24)	11.0 %
BC-vision-only ^B	0.089 $\pm$ 0.018	54.7 $\pm$ 3.4 %	61.1 $\pm$ 13.4 %	3.30 $\pm$ 0.36 m (18)	9.2 $\pm$ 1.8 %

Δ `n_cond` = 4: cond-IAE unreliable; the low all-IAE is a short-survival artifact. $\Delta\Delta$ PPO-from-pixels is collapse-prone, not uniformly collapsed: 2 of 3 seeds survive 0/30 past 250 steps (`n_cond` = 0, cond-IAE undefined; their low all-IAE 0.70–0.87 m is a pure short-survival artifact, crashing $\sim$ 75 steps), while 1 seed reaches Tier-1 46.7 % (cond-IAE 2.83 m). The cond-IAE shown is that single surviving seed (1/3). ^B P1 baseline (this work): **mean $\pm$ std over 3 seeds {0,1,2}** (population std, matching Table 2); the bracketed [..] on the other rows are bootstrap 95 % CIs over a single checkpoint's episodes. BC-vision-only = no IMU, no flow (plain vision→action MLP); PPO-from-pixels = end-to-end RL from pixels (1 M steps, CTBR).

The two P1 baselines (3 seeds each) sit within the flow-policy band and reinforce the diagnosis. A plain vision→action regressor (BC-vision-only: no IMU, no flow) already reaches **9.2 $\pm$ 1.8 % of oracle at Tier-1 61.1 $\pm$ 13.4 %** — *on par with* the IMU-fused flow policies v5_BC (12.8 %, Tier-1 70 %) and Joint_E2E_v5 (11.0 %, Tier-1 80 %), well within the wide across-seed spread. Flow modelling and IMU fusion thus buy little in survival on this task: **policy sophistication is not**

the bottleneck. The other baseline, PPO-from-pixels, is the collapse-prone case dissected above; here it doubles as a clean short-survival-artifact demonstration. On its two collapsed seeds the deceptively low *all*-IAE (0.70–0.87 m) is a textbook instance of what the §4 protocol is built to expose — they "look precise" only because they crash (~75 steps) before they can drift — whereas averaging over seeds raises the *all*-IAE to 1.41 ± 0.89 m once the surviving seed's drift is counted. Both baselines stay far below the 17 % ceiling; none is deployable.

One training seed is not a safe leaderboard row (P1 baselines, 3 seeds each)

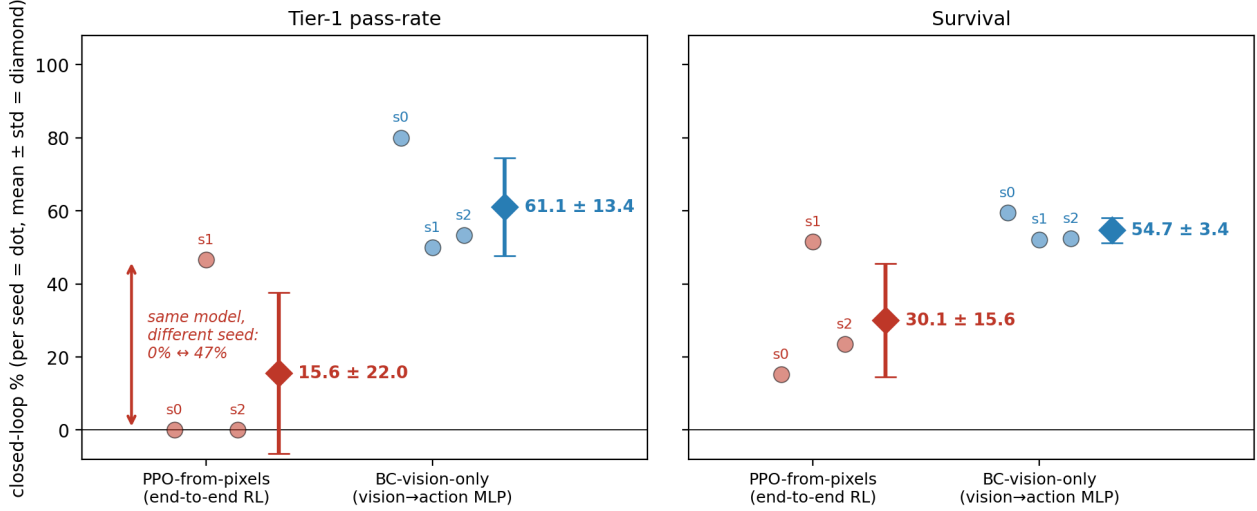


Figure 3. Per-seed Tier-1 pass-rate (left) and survival (right) for the two P1 baselines, three training seeds each (dots, labelled s0–s2), with the across-seed mean $\pm$ std (diamond). PPO-from-pixels swings from 0 % (seeds 0, 2 — would read as "uniformly collapsed") to 47 % (seed 1) on Tier-1 from the training seed alone (15.6 ± 22.0 %), so any single-seed leaderboard row simultaneously over- and under-states it; BC-vision-only is comparatively stable (61.1 ± 13.4 %). This seed sensitivity is why every retrained model in this work is reported as a three-seed mean $\pm$ std (`scripts/make_paper_figures.py`, `evaluation_results/baselines_frozen_seeds_aggregate.json`).

5. The Core Ablation: Dispersive $\times$ E2E (2 $\times$ 2, 3 seeds)

We vary two factors, holding everything else fixed (H4-transfer init, 500 hover + 500 recovery demos, 80 epochs, lr $1e-4$, batch 256, task-conditioned):

- **Dispersive:** the official-code-faithful InfoNCE-L2 term (§3) at $\lambda=0.5$ (ON) vs 0.0 (OFF), on the `flow_net` mid-block.
- **E2E:** vision encoder trainable (ON) vs frozen at the transferred init (OFF).

Design and power. With three seeds per cell and a pooled across-seed Tier-1 std of ~ 6.3 pp, this design is powered to detect only large effects (≥ 6 pp Tier-1); a smaller true effect of Dispersive cannot be excluded. We therefore report a null in the sense of *no effect above seed noise*, not a proof of exact zero, and judge support by whether D1E1 exceeds DoE1 by more than the pooled std.

What the frozen-encoder cell tests. Because the term lands on `flow_net` mid-features, it trains the generative network *even with a frozen encoder*. The frozen-encoder cell (Eo) is therefore not a trivial no-op: it isolates Dispersive's effect with the *encoder* held fixed but the *flow network* still adapting. The decisive hypothesis test is **D1E1 vs DoE1** (Dispersive ON vs OFF, both E2E).

Table 2 — 2 $\times$ 2 ablation, faithful Dispersive (`evaluate_p2_ablation.py`, 3 seeds, frozen protocol; `evaluation_results/p2f_ablation_leaderboard.json`).

Cell	Disp / E2E	Tier-1 (mean $\pm$ std)	Survival (mean $\pm$ std)	cond-IAE	%Oracle
DoEo	OFF / frozen	87.8 ± 3.1	66.1 ± 3.7	2.93 m	13.4 %
D1Eo	ON / frozen	74.4 ± 8.7	60.4 ± 1.6	2.81 m	12.9 %
DoE1	OFF / E2E	92.2 ± 3.1	65.0 ± 2.8	2.91 m	12.4 %
D1E1	ON / E2E	90.0 ± 5.4	62.9 ± 2.4	2.89 m	12.2 %

Faithful Dispersive $\times$ E2E (2 $\times$ 2, 3 seeds): no Dispersive effect above seed noise

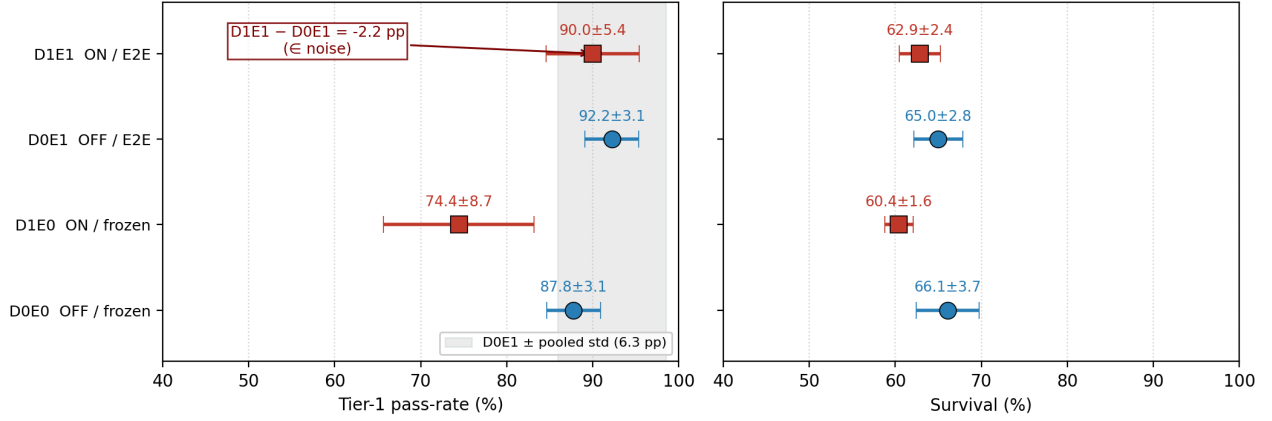


Figure 5. The 2 $\times$ 2 ablation as a forest plot (faithful Dispersive, 3 seeds, mean $\pm$ across-seed std). *Left:* Tier-1; the grey band is DoE1 $\pm$ the pooled across-seed std (6.3 pp). The decisive D1E1 (ON/E2E) sits inside it — **no Dispersive effect above seed noise** (-2.2 pp). The one cell that moves is D1E0 (ON/frozen, 74.4 ± 8.7), where Dispersive-on-frozen lowers the mean and inflates the variance. *Right:* survival is flat across all four cells. (scripts/make_paper_figures.py $\rightarrow$ evaluation_results/p2f_ablation_leaderboard.json.)

Findings. - Dispersive is not supported (Figure 5). D1E1 vs DoE1: Tier-1 -2.2 pp inside the 6.3 pp pooled across-seed std; survival -2.1 pp; conditional precision unchanged (2.89 vs 2.91 m). The decision rule (supported iff D1E1 beats DoE1 by more than the pooled std) returns **NOT supported**. The conclusion holds under the off-path, low-weight variant of §6.1 as well ($+1.1$ pp), so it does not hinge on placement or weight. - **The frozen-encoder condition is an informative control, not a no-op.** Under the faithful mid-block placement the encoder-frozen runs *do* train `flow_net`, so the checkpoints differ (`p2f_D1E0_s*` $\neq$ `p2f_D0E0_s*`, MD5 distinct across all three seeds). The frozen row is not inert: Dispersive-on-frozen **destabilises** control (mean Tier-1 $87.8 \rightarrow 74.4$, variance $3.1 \rightarrow 8.7$ pp) — with no trainable encoder to co-adapt, repelling the mid-features pushes the mean down and the variance up rather than helping. - **E2E is the only (small) Tier-1 mover** (DoE0 $\rightarrow$ DoE1, $+4.4$ pp), and even it neither extends survival nor improves precision (§6.2).

All four cells beat the prior frontier Joint_E2E_v5 (Tier-1 80 %), but §6.2 shows that gain is the *recipe*, not Dispersive. Precision is unmoved (cond-IAE 2.7–3.2 m, ~ 12 –13 % oracle): **nothing is deployable**.

6. Diagnosis: Which Bottleneck?

Three candidate explanations for the flat result: representation (the thing Dispersive targets), data coverage, or sensing. We test the first two with observation-only measurements (no retraining), and the third with both a renderer gate and a positive-control **intervention** that hands the policy the very signal it is presumed to lack (§6.3).

6.1 Representation: Dispersive games its objective; survival $\perp$ rank

Placement note. We measure feature geometry at **both** placements. We start with the off-path `vis_pooled` placement (Table 3; legacy P2 checkpoints): because `vis_pooled` feeds only the auxiliary state head, it cleanly isolates how a scale-sensitive dispersion term games its objective off the action path. We then confirm the same pathology at the faithful `flow_net` mid-block, where the official term actually acts (Table 4; the p2f checkpoints of §5). One placement-dependent detail: off-path, a frozen encoder zeroes the gradient (so D1E0 $\equiv$ DoE0, the legacy no-op); at the faithful placement the term trains `flow_net` even when frozen (§5), so D1E0 $\neq$ DoE0.

We push a fixed 4000-image hover+recovery batch through each checkpoint's encoder and measure the geometry of `vis_pooled` (D=256). (`measure_feature_collapse.py`, legacy P2 checkpoints.)

Table 3 — feature geometry (mean over 3 seeds).

Cell	eff_rank	dims for 99% var	feat_norm	L_disp	mean cos
DoE0 / D1E0 (frozen)	30.3	91	1.80	-0.81	0.071
DoE1 (E2E)	9.0	45	11.42	-1.23	0.956
D1E1 (E2E + Disp)	2.0	2	3281	-8.14	0.013

Three things follow. **(i)** The frozen no-op is confirmed at the feature level ($D1E0 \equiv D0E0$ to 4 d.p.). **(ii)** Collapse is *real*: naive E2E ($D0E1$) drops effective rank 30→9 and drives mean pairwise cosine to 0.96 (features nearly colinear) — there *was* something for Dispersive to fix. **(iii)** Dispersive does not fix it; it **games its objective**. It minimises $\mathcal{L}_{disp}$ ($-1.23 \rightarrow -8.14$) and inflates pairwise distance $\sim 1270\times$, but does so purely by inflating the feature norm $\sim 287\times$ ($11.4 \rightarrow 3281$); the *intrinsic* dimensionality gets **worse** (effective rank 9→2; 99.8 % of variance on 2 dims). The "healthy-looking" low cosine (0.013) is a 2-D-at-huge-radius artifact, not high- dimensional spread. Because `vis_pooled` feeds only the auxiliary head (§3), the cheapest way to satisfy the distance objective is to inflate the pooled-specific projection, leaving the action-relevant spatial pathway comparatively untouched.

The decisive plot is rank vs survival (Figure 1). Effective rank swings $15\times$ (30→9→2) while closed-loop survival stays flat ($66.1 \rightarrow 65.0 \rightarrow 65.0$ %, within ~ 3 pp std). **Survival is decoupled from `vis_pooled` rank.** Within this off-path probe, neither preventing collapse ($D0E0$ rank 30) nor worsening it ($D1E1$ rank 2) moves the closed-loop outcome. Together with the faithful on-path null (§5), this is the paper's strongest evidence that, *for this task*, representation collapse is not the binding constraint. The same pathology and decoupling hold at the faithful placement (Table 4, below).

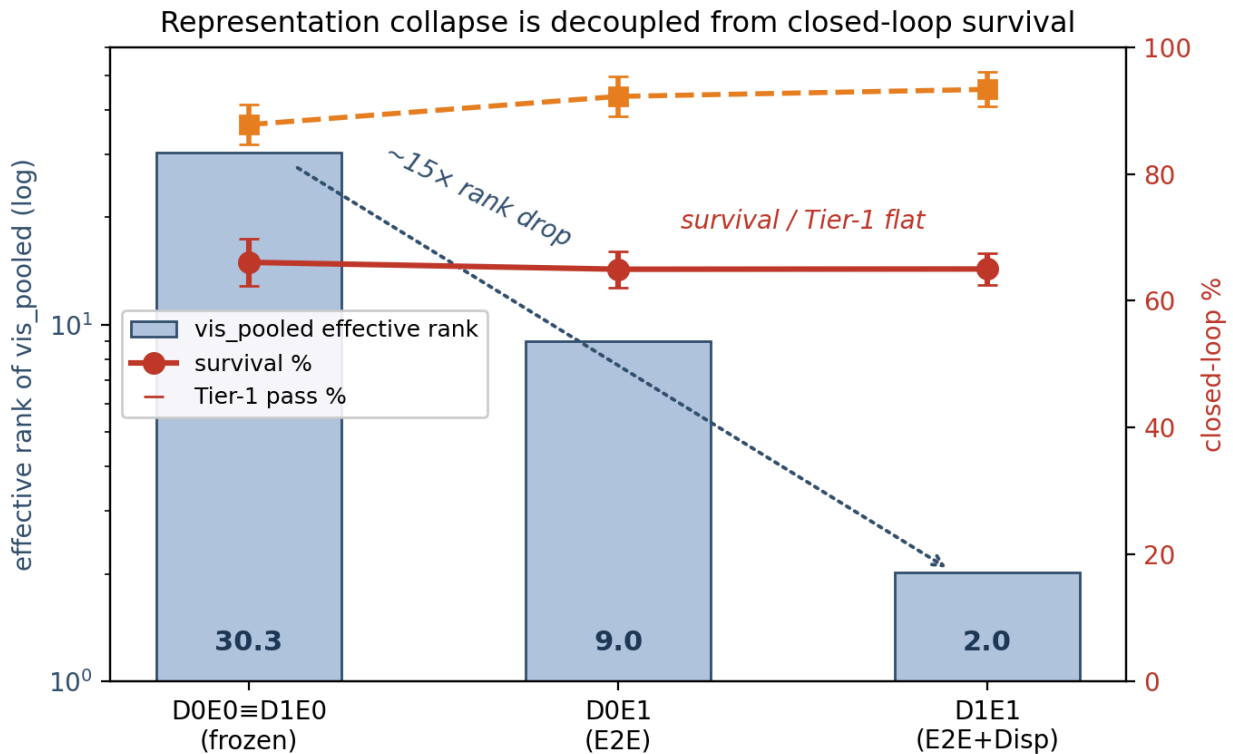


Figure 1. `vis_pooled` effective rank (bars, log axis) collapses $\sim 15\times$ across the 2×2 cells (30.3 frozen $\rightarrow$ 9.0 E2E $\rightarrow$ 2.0 E2E+Dispersive), while closed-loop survival and Tier-1 pass-rate (lines, right axis; error bars = across-seed std) stay flat. Survival is decoupled from the representation geometry Dispersive targets.

The same pathology holds at the faithful placement (Table 4). The off-path probe above cannot say whether the faithful `flow_net` mid-features — where the official term actually acts — behave the same way. Measuring them directly (same fixed 4000-sample batch, now of (image, IMU, action, timestep $\mathbf{t}$) so the mid-features are well defined; `measure_feature_collapse_flowmid.py`) returns the same verdict, more sharply. Dispersive ON drives the InfoNCE-L2 objective to its subset floor ($\log(1/2048) = -7.62$, identical across all three seeds) — it fully *wins* its objective — but does so purely by inflating the mid-feature norm $\sim 9\times$ ($9.4 \rightarrow 84$) while the effective rank **collapses** ($221 \rightarrow 36$ with a trainable encoder, $249 \rightarrow 37$ frozen; dims-for-99 %-variance $683 \rightarrow 177$). Exactly as off-path, the term games a scale-sensitive objective by norm growth and *worsens* intrinsic rank. Two further points: (i) naive E2E barely dents the mid-feature rank ($D0E0 \rightarrow D0E1$, $249 \rightarrow 221$) — unlike `vis_pooled` (30→9), the action-path mid-features do not strongly collapse on their own, so there is little for Dispersive to "fix"; (ii) the frozen-Dispersive cell ($D1E0$) shows the *same* norm blow-up and rank collapse as $D1E1$, a plausible mechanism for its closed-loop destabilisation (§5) — repelling the velocity field's own mid-features inflates them $\sim 9\times$ with no encoder free to compensate. Across all four cells the mid-feature rank swings $\sim 7\times$ ($249 \rightarrow 36$) while survival stays within ~ 3 pp (§5): the rank-vs-survival decoupling holds at the faithful placement too.

Table 4 — faithful `flow_net` mid-block geometry (D=1024, mean±std over 3 seeds; `measure_feature_collapse_flowmid.py` $\rightarrow$ `evaluation_results/p2f_feature_collapse_flowmid.json`).

Cell	eff_rank	dims for 99% var	feat_norm	disp_infonce	mean cos
DoEo (OFF / frozen)	248.8 ± 9.0	722	9.35	-0.283	0.174
D1Eo (ON / frozen)	36.7 ± 0.5	189	83.8	-7.624	0.058
DoE1 (OFF / E2E)	221.4 ± 16.2	683	9.49	-0.293	0.163
D1E1 (ON / E2E)	35.9 ± 0.6	177	84.8	-7.624	0.056

6.2 The only survival mover is the recipe, not Dispersive (or E2E)

Decomposing the gain over the prior frontier as controlled steps (`experiment_report_survival_movers.md`):

Step	Δ Tier-1	Δ Survival	cond-IAE
prior frontier Joint_E2E_v5 → recipe (frozen cell DoEo)	+7.8 pp	+3.9 pp	2.93 m
+ E2E (DoEo → DoE1)	+4.4 pp	-1.1 pp	2.91 m
+ Dispersive, faithful (DoE1 → D1E1)	-2.2 pp (€ noise)	-2.1 pp	2.89 m

(The DoEo/DoE1 cells are Dispersive-OFF, so the +Dispersive step is the only one that depends on the regulariser: under the official configuration it is -2.2 pp, and under the off-path low-weight variant of §6.1 it is +1.1 pp — both inside the across-seed std.)

The dominant lift is the **recipe** (H4-transfer init + task-conditioning + recovery mix), and it holds with a **frozen** encoder — so it is not from E2E. E2E adds a small Tier-1-only bump that neither extends survival nor improves precision. Dispersive adds nothing (slightly negative under the official configuration, € noise). **Mover ranking: recipe » E2E » Dispersive ≈ 0.**

6.3 Precision is coverage/teacher-competence-limited, not sensing-limited

cond-IAE is ~2.8 m (~13 % oracle) for *every* configuration. Why won't precision move?

Coverage probe (`measure_ood_coverage.py`). Comparing position-error magnitude in the BC data vs the closed-loop steady state: training is overwhelmingly at-target (95.8 % of timesteps within 0.2 m; p99 = 0.62 m), while the surviving policy spends ~90 % of its steady state beyond 1 m and 46 % beyond 3 m. **97.3 %** of steady-state samples lie above the training p99; **25 %** above the training maximum (zero-coverage). The policy lives where it has ~no labels — a real coverage gap, *consistent* with either a data limit or a sensing limit.

Is the coverage gap data-fixable, and does the FPV even encode range? Three observation-only checks bear on it: 1. **Widening the recovery init does not change coverage.** The env anchors `target = init_pos` for hover, so a wider `--pos-range` just relocates the hover point; a freshly collected ±3 m dataset has *identical* position-error coverage to the ±1 m one (both 0.4 % of timesteps beyond 1 m). 2. **The expert itself cannot recover from > 2 m offset** (`target ≠ init`): 1 m and 2 m recover to 0.066 m (20/20), but 3 m crashes (0/20). Even a correct offset collection tops out at ~2 m — below the precision-limiting regime. 3. **The FPV image does not encode metric range** (`measure_image_distance_info.py` ; **Figure 2**). The only range cue is the target crosshair size, `size = max(2, min(6, [6/(d+0.5)] + dr))` . With domain randomisation, adjacent-distance separability is $d' < 0.2$ everywhere, and a linear decode of distance from the image has $R^2 = 0.41$ near (< 1 m) but only **0.12 far (≥ 1.5 m)**. With DR off (noiseless ceiling) the size **saturates at 2 px for all $d \geq 2$ m**, so renders at 2.0, 2.5, and 3.0 m are **byte-identical** ($d' = 0$). The crosshair *position* encodes only normalised direction (range-invariant).

The policy's 2.83 m steady drift sits exactly where the observation carries **no recoverable metric range** (Figure 2). One might conclude that precision is *information-gated by the observation model*. That inference does not follow: a measurement that the FPV does not encode range does not establish that the missing range is the *binding* cause of imprecision. We test that implication directly with a renderer gate and a positive-control intervention, and **both refute it**.

Measurement: the FPV cannot encode metric range where the policy operates (but see Fig. 4 — this is a fixable artifact, and range is not what gates precision)

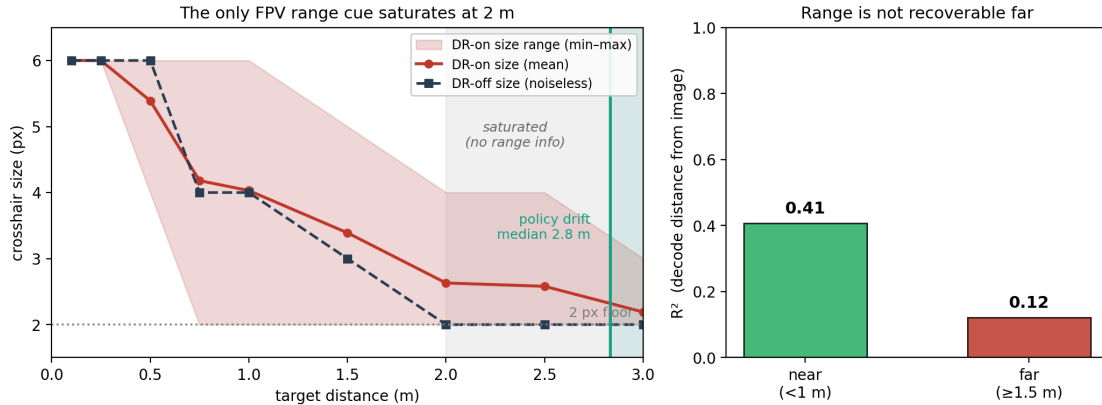


Figure 2. *Left:* the only FPV range cue (target crosshair size) saturates at the 2 px floor beyond ~2 m — under domain randomisation (red band = min–max) the size is uninformative even nearer, and with DR off (dashed) renders at 2.0/2.5/3.0 m are byte-identical. The policy’s steady-state drift (green, median 2.83 m) sits inside the saturated region. *Right:* a linear decode of distance from the image succeeds near ($R^2=0.41$, <1 m) but fails far ($R^2=0.12$, ≥ 1.5 m). This is a *measurement* of the observation, not the binding cause of imprecision (Figure 4).

The information loss is a renderer artifact, not the pixel count (gate). Before treating "richer sensing" as the fix, we ask whether a more capable monocular renderer would even *carry* the far-range information (`measure_higher_res_gate.py` ; **Figure 4, left**). Decoding distance from the image across $\{64, 128, 256\}$ px $\times$ {production saturating crosshair, perspective non-saturating target}, the saturating crosshair’s far-range R^2 is ≈ 0 at *every* resolution (-0.01 / 0.11 / 0.04) — raising resolution alone does nothing — whereas a perspective (optical-expansion) target lifts far R^2 to **0.42 at the same 64 px** (0.50 / 0.45 at 128 / 256 px). So the §6.3 measurement is partly an artifact of this synthetic renderer’s quantised crosshair, fixable without any resolution change; even so, the perspective target’s far R^2 (0.45) $\ll$ its near R^2 (0.88), so a richer sensor would *improve*, not solve, far-range precision.

Even the oracle range cue does not move precision (intervention). The decisive test supplies the missing channel directly: we fold the metric body-frame position error the FPV cannot encode (`states[:, :3]` , computed from the existing data — no re-collection) into the policy’s conditioning, retrain the DoE1 frontier recipe unchanged, and re-eval under the frozen protocol (3 seeds; control = the no-cue DoE1; **Figure 4, right**). Handed the *oracle* metric range (scalar `||pos-err||`, $\sigma=0$), cond-IAE moves only $2.91 \rightarrow$ **2.43 m** — still $\sim 36\times$ the 0.068 m state oracle, and bought with -6.7 pp survival / -13 pp Tier-1. A realisable sensor ($\sigma=0.15$ m) erases even that (2.81 m $\approx$ control). The *richer* full 3-D position cue is actively harmful: it **collapses survival across all three seeds** (Tier-1 $92 \rightarrow 7\%$, surviving to the 250-step threshold in $\approx 2/30$ episodes), so its conditional precision is an artifact, not a win. Supplying range tells the drone *how far off* it is, but not *what to do* about it — and no demonstration in the 1–3 m band teaches that.

Verdict: precision is not sensing-gated. Range information — even oracle, even full position — does not restore precision, and a richer cue harms survival. The binding constraint is the absence of a learned far-range *recovery behaviour* in the 1–3 m band: the BC data has no labels there (coverage gap, above) and the teacher cannot generate them (the expert itself crashes from > 2 m, check 2). Moving precision would require a *competent far-range teacher* to generate 1–3 m coverage — not a better sensor, not a better policy over the existing data, and not representation regularisation. A wider-init BC retrain is therefore still predicted to leave cond-IAE at ~ 2.8 m and is not pursued.

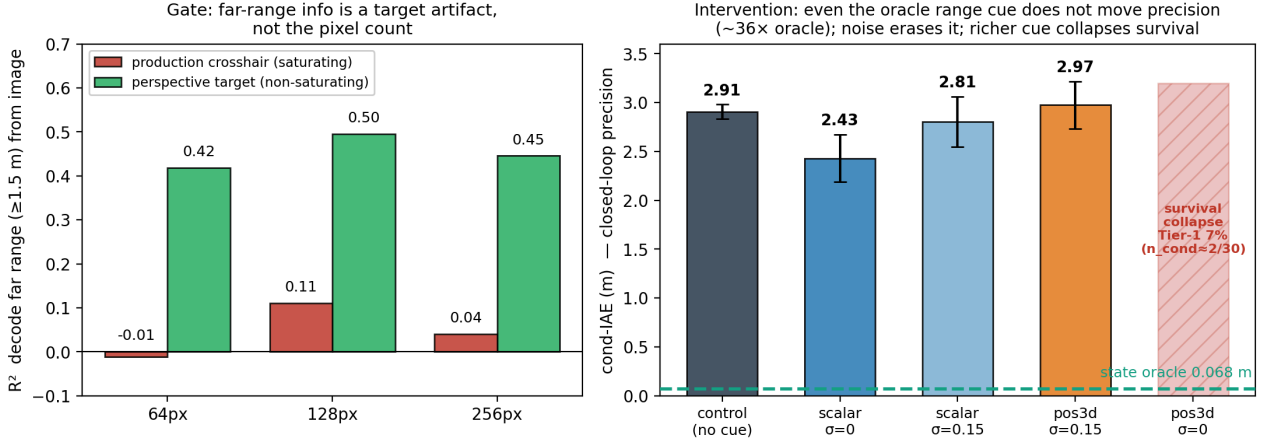


Figure 4. *Left (gate):* far-range (≥ 1.5 m) image $\rightarrow$ distance R^2 for three resolutions $\times$ two target renderers. The saturating production crosshair is uninformative at every resolution (raising pixels does nothing); a non-saturating perspective target restores $\sim$ half the far-range information already at 64 px — the loss is a target artifact, not the pixel count. *Right (intervention):* closed-loop precision (cond-IAE, 3 seeds) when the policy is *handed* the metric position error the FPV lacks. Even the oracle scalar range ($\sigma=0$) leaves precision $\sim 36\times$ the state oracle (green dashed, 0.068 m); sensor noise erases the gain; the richer 3-D cue ($\sigma=0$) collapses survival (hatched; conditional IAE unreliable). Precision is coverage/teacher-competence-gated, not sensing-gated.

7. Discussion

What generalises. (a) The frozen protocol and the short-survival artifact it fixes are reusable for any survival-vs-precision visual control benchmark. (b) The "objective-gaming" pathology of a *scale-sensitive* dispersion regulariser on an *off-path* feature is a general caution: a distance-maximising contrastive term can be trivially satisfied by norm inflation, *worsening* intrinsic rank; one should regularise the feature the policy actually conditions on, and use a scale-invariant criterion. The faithful on-path placement (§5) closes that obvious escape — there is no auxiliary-only projection to inflate — yet it still does not help (no gain when the encoder is trainable; mildly destabilising when frozen), so neither placement rescues closed-loop control. (c) The rank-vs-survival decoupling argues that "prevent representation collapse" is not automatically the right lever for closed-loop control — the binding constraint must be located empirically.

What is specific. The precision diagnosis (§6.3) is specific to a setting whose privileged teacher saturates inside the operating regime: our state-based PPO oracle itself crashes from > 2 m offset, so it cannot label the 1–3 m band where the policy drifts. A task with a teacher competent over the full error range — or a renderer whose distance cue is not the quantised crosshair we found to be artifactually saturating — could place the bottleneck elsewhere.

Constructive implication. Improving metric precision here is neither a representation nor a sensing problem: our intervention shows that even the *oracle* range, handed directly to the policy, barely moves precision and a richer cue collapses survival (§6.3). The lever is the *teacher and its coverage* — a controller that recovers from multi-metre offsets, generating 1–3 m demonstrations to imitate. A better sensor would help only once such recovery behaviour exists to be conditioned on; absent it, higher resolution, stereo/depth, representation regularisation, and wider naive BC coverage are all predicted not to move the ~ 2.8 m floor.

8. Limitations

- **Simulation only;** no real-robot validation. The renderer is synthetic and its distance encoding is a design choice (a deliberately information-poor monocular cue) — though the §6.3 intervention controls for this by handing the policy the oracle range directly, so the precision conclusion does not hinge on the renderer's information content.
- The precision verdict is **conditional on the teacher:** our state-based PPO oracle itself crashes from > 2 m offset, so "coverage/teacher-competence-gated" is established for *this* teacher; a controller competent over the full error range is the untested lever and the recommended next step.
- **One task** (hover) and one policy family (flow matching with IMU-vision cross-attention). The headline Dispersive null result is established at the **official flow_net mid-block placement** ($\lambda=0.5$, $\tau=0.5$, $/d$); we additionally probe the off-

path `vis_pooled` placement mechanistically (§6.1). We did not exhaustively sweep λ , τ , or every candidate intermediate block, or alternative dispersion criteria — though §6.1–§6.2 explain mechanistically why those are unlikely to rescue it here.

- **Statistical power.** Three seeds per cell with a pooled across-seed Tier-1 std of ~ 6.3 pp powers this design to detect only large effects (≥ 6 pp Tier-1). The result is a null *in the sense of no effect above seed noise*; a smaller true effect of Dispersive — and the directional, high-variance frozen-encoder degradation — cannot be sharply resolved at this seed count.
- $T_{\text{obs}} = 2$ frames could in principle yield range via motion parallax; we argue (and the flat empirical precision across all configs supports) that the 28 ms baseline and slow drift make this far weaker than the static size channel.

9. Conclusion

On a vision-based quadrotor hover task, under a frozen, artifact-robust protocol with a measured oracle and an **official-code-faithful** Dispersive implementation (InfoNCE-L2 on the `flow_net` mid-block, $\lambda=0.5$, τ/d), **Dispersive Loss does not improve survival or precision** above seed noise (~ 2.2 pp Tier-1, $\in$ noise). Even with a frozen encoder — where the faithful term still trains the generative network — it does not help and is mildly destabilising, so the frozen-encoder condition is an informative control rather than a no-op. At **both** the faithful and an off-path placement the mechanism does not even cure the collapse it targets — it games its scale-sensitive objective and worsens intrinsic rank — and closed-loop survival is decoupled from that representation geometry entirely. The real levers are elsewhere: a transfer/conditioning/recovery recipe accounts for the survival gains, and **task precision is capped by the teacher's incompetence in the operating regime**, not by sensing: a positive-control intervention that hands the policy the oracle metric position error barely moves precision ($\sim 36\times$ the state oracle) and a richer cue collapses survival, and the FPV's apparent range-blindness is itself a fixable renderer artifact. Neither representation collapse nor the sensing channel is the binding constraint for this task; the missing far-range recovery behaviour is. We release the protocol, the ablation, the intervention, and all diagnostics so the result and its diagnosis are reproducible.

Reproducibility / Artifacts

Component	Script	Artifact
Frozen protocol + measured oracle	<code>scripts/evaluate_frozen_p0.py</code>	<code>evaluation_results/frozen_p0_leaderboard.json</code>
§4 P1 baselines (BC-vision-only, PPO-from-pixels), 3 seeds	<code>scripts/train_bc_vision_only.py</code> , <code>scripts/train_ppo_from_pixels.py</code> (+ <code>models/ppo_pixel.py</code> , <code>configs/ppo_from_pixels.yaml</code>), <code>scripts/evaluate_baselines_frozen.py</code> ; seed driver <code>scripts/run_baseline_seeds_1_2.sh</code>	seed 0: <code>evaluation_results/baselines_frozen_leaderboard.json</code> , ... <code>_ppopx.json</code> ; seeds 1–2: ... <code>{bc,ppopx}_s{1,2}.json</code> ; aggregate (mean±std): ... <code>_baselines_frozen_seeds_aggregate.json</code>
§5 faithful 2×2 (P2f, headline)	<code>scripts/run_p2_ablation.py --faithful</code> , <code>scripts/evaluate_p2_ablation.py</code>	<code>evaluation_results/p2f_ablation_{manifest,leaderboard}.json</code> — <code>docs/experiment_report_faithful_dispersive.md</code>
legacy 2×2 ablation (off-path, §6.1 probe)	<code>scripts/run_p2_ablation.py</code> , <code>scripts/evaluate_p2_ablation.py</code>	<code>evaluation_results/p2_ablation_{manifest,leaderboard}.json</code>
§6.1 feature geometry — off-path <code>vis_pooled</code> (Table 3)	<code>scripts/measure_feature_collapse.py</code>	<code>evaluation_results/p2_feature_collapse.json</code> — <code>docs/experiment_report_feature_collapse.md</code>
§6.1 feature geometry — faithful <code>flow_net</code> mid-block (Table 4)	<code>scripts/measure_feature_collapse_flowmid.py</code>	<code>evaluation_results/p2f_feature_collapse_flowmid.json</code>
§6.2 survival movers	—	<code>docs/experiment_report_survival_movers.md</code>
§6.3 coverage probe	<code>scripts/measure_ood_coverage.py</code>	<code>evaluation_results/p3b_ood_coverage.json</code> — <code>docs/experiment_report_ood_coverage.md</code>
§6.3 image-distance info	<code>scripts/measure_image_distance_info.py</code>	<code>evaluation_results/p3b_image_distance_info{,_nodr}.json</code> — <code>docs/experiment_report_image_distance_info.md</code>
§6.3 higher-res gate (Fig 4 left)	<code>scripts/measure_higher_res_gate.py</code>	<code>evaluation_results/p3b_higher_res_gate.json</code> — <code>docs/experiment_report_sensing_ablation.md</code>
§6.3 range-cue intervention, 3 seeds (Fig 4 right)	<code>scripts/run_p3b_rangecue.py</code> (→ <code>train_flow_v5.py --range-cue</code> , <code>evaluate_frozen_p0.py --cue-noise</code>)	<code>evaluation_results/p3b_rc_{clean,noised}{,_s12}_frozen.json</code> — <code>docs/experiment_report_sensing_ablation.md</code>

Frozen protocol: 30 episodes, base seed 12345, $\sigma = 2.0$ exp-decay composite, paired init, conditional-IAE over episodes surviving $\geq 250/500$ steps, bootstrap 95% CI, measured PPO oracle 0.9668.

References

Sources retrieved via NotebookLM (notebook *Generative RL & Flow Policy Research*, 2026-06-19). All author lists, venues, and identifiers were verified against the publisher of record (arXiv / official proceedings) on 2026-06-19; no entries remain unverified.

[1] J. Ho, A. Jain, P. Abbeel. "Denoising Diffusion Probabilistic Models." *NeurIPS*, 2020.

[2] J. Song, C. Meng, S. Ermon. "Denoising Diffusion Implicit Models." *ICLR*, 2021.

- [3] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, S. Song. "Diffusion Policy: Visuomotor Policy Learning via Action Diffusion." *Robotics: Science and Systems (RSS)*, 2023. (Extended version: *Int. J. Robotics Research*, 2024.)
- [4] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, M. Le. "Flow Matching for Generative Modeling." *ICLR*, 2023.
- [5] X. Liu, C. Gong, Q. Liu. "Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow." *ICLR*, 2023.
- [6] K. Black, N. Brown, D. Driess, et al. (Physical Intelligence). " π_0 : A Vision-Language-Action Flow Model for General Robot Control." arXiv:2410.24164, 2024.
- [7] M. Braun, N. Jaquier, L. Rozo, T. Asfour. "Riemannian Flow Matching Policy for Robot Motion Learning." *IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, pp. 5144–5151, 2024.
- [8] A. Z. Ren, J. Lidard, L. L. Ankile, A. Simeonov, P. Agrawal, A. Majumdar, B. Burchfiel, H. Dai, M. Simchowitz. "Diffusion Policy Policy Optimization (DPPO)." *ICLR*, 2025. arXiv:2409.00588.
- [9] T. Zhang, C. Yu, S. Su, Y. Wang. "ReinFlow: Fine-tuning Flow Matching Policy with Online Reinforcement Learning." arXiv:2505.22094, 2025.
- [10] K. Black, M. Janner, Y. Du, I. Kostrikov, S. Levine. "Training Diffusion Models with Reinforcement Learning (DDPO)." arXiv:2305.13301, 2023.
- [11] Y. Fan, O. Watkins, Y. Du, et al. "DPOK: Reinforcement Learning for Fine-tuning Text-to-Image Diffusion Models." *NeurIPS*, 2023. arXiv:2305.16381.
- [12] M. Psenka, A. Escontrela, P. Abbeel, Y. Ma. "Learning a Diffusion Model Policy from Rewards via Q-Score Matching (QSM)." *ICML*, 2024. arXiv:2312.11752.
- [13] R. Wang, K. He. "Diffuse and Disperse: Image Generation with Representation Regularization." arXiv:2506.09027, 2025. (Submitted to *ICLR* 2026.)
- [14] G. Zou, W. Li, H. Wu, Y. Qian, Y. Wang, H. Wang. "D²PPO: Diffusion Policy Policy Optimization with Dispersive Loss." *AAAI*, 2026. arXiv:2508.02644.
- [15] S. Yu, S. Kwak, H. Jang, J. Jeong, J. Huang, J. Shin, S. Xie. "Representation Alignment for Generation: Training Diffusion Transformers Is Easier Than You Think (REPA)." *ICLR (Oral)*, 2025. arXiv:2410.06940.
- [16] A. Bardes, J. Ponce, Y. LeCun. "VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning." *ICLR*, 2022. arXiv:2105.04906.
- [17] J. Zbontar, L. Jing, I. Misra, Y. LeCun, S. Deny. "Barlow Twins: Self-Supervised Learning via Redundancy Reduction." *ICML*, 2021. arXiv:2103.03230.
- [18] G. Zou, H. Wang, H. Wu, Y. Qian, Y. Wang, W. Li. "DM1: MeanFlow with Dispersive Regularization for 1-Step Robotic Manipulation." arXiv:2510.07865, 2025.
- [19] J. Sheng, Z. Wang, P. Li, M. Liu. "MP1: MeanFlow Tames Policy Learning in 1-step for Robotic Manipulation." arXiv:2507.10543, 2025.
- [20] A. Loquercio, E. Kaufmann, R. Ranftl, M. Müller, V. Koltun, D. Scaramuzza. "Learning High-Speed Flight in the Wild." *Science Robotics*, 6(59), 2021.
- [21] E. Kaufmann, L. Bauersfeld, A. Loquercio, M. Müller, V. Koltun, D. Scaramuzza. "Champion-Level Drone Racing using Deep Reinforcement Learning." *Nature*, 620, 2023.
- [22] L. Jing, P. Vincent, Y. LeCun, Y. Tian. "Understanding Dimensional Collapse in Contrastive Self-Supervised Learning." *ICLR*, 2022. arXiv:2110.09348.
- [23] G. Zou, H. Wang, H. Wu, Y. Qian, Y. Wang, W. Li. "One Step is Enough: Dispersive MeanFlow Policy Optimization (DMPO)." arXiv:2601.20701, 2026.
- [24] E. Kaufmann, M. Gehrig, P. Foehn, R. Ranftl, A. Dosovitskiy, V. Koltun, D. Scaramuzza. "Deep Drone Racing: Learning Agile Flight in Dynamic Environments." *Conf. on Robot Learning (CoRL)*, 2018. arXiv:1806.08548.
- [25] E. Kaufmann, A. Loquercio, R. Ranftl, M. Müller, V. Koltun, D. Scaramuzza. "Deep Drone Acrobatics." *Robotics: Science and Systems (RSS)*, 2020.
- [26] S. Ross, G. J. Gordon, J. A. Bagnell. "A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning (DAgger)." *AISTATS*, 2011. arXiv:1011.0686.
- [27] M. Schneider, R. Krug, N. Vaskevicius, L. Palmieri, J. Boedecker. "The Surprising Ineffectiveness of Pre-Trained Visual Representations for Model-Based Reinforcement Learning." *NeurIPS*, 2024. arXiv:2411.10175.

- [28] P. Henderson, R. Islam, P. Bachman, J. Pineau, D. Precup, D. Meger. "Deep Reinforcement Learning that Matters." *AAAI*, 2018. arXiv:1709.06560.
- [29] R. Agarwal, M. Schwarzer, P. S. Castro, A. C. Courville, M. G. Bellemare. "Deep Reinforcement Learning at the Edge of the Statistical Precipice." *NeurIPS*, 2021. arXiv:2108.13264.
- [30] A. Patterson, S. Neumann, M. White, A. White. "Empirical Design in Reinforcement Learning." *Journal of Machine Learning Research*, 25, 2024. arXiv:2304.01315.